

FVA and balance sheet

Pre-default costs vs post-default windfalls

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Cass Financial Engineering Workshop
London 5th Nov 2014

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Overview

- 1 Introduction
- 2 FVA in a nutshell
- 3 CVA desks vs Funding desks - practical setups
- 4 Replication and funding strategies
- 5 Example: Set-offs
- 6 Different funding strategies
- 7 Risk neutral pricing and balance sheet effects
- 8 Accounting for FVA
- 9 Conclusions

1 Introduction

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Bloomberg

It Cost JPMorgan \$1.5 Billion to Value Its Derivatives



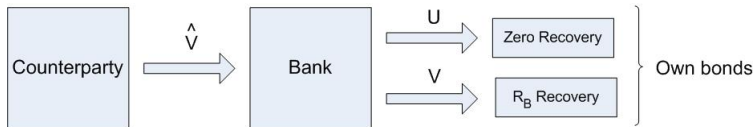
Nomura books ¥10 billion FVA loss

FTAlphaville

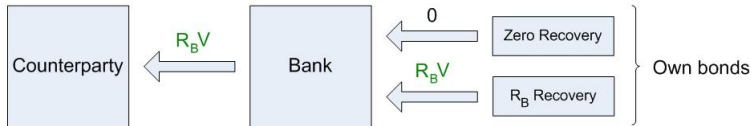
DVA, CVA and FVAaaaaaaargh!

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FVA in a nutshell: Derivatives liability



Upon default:



Perfectly hedged!

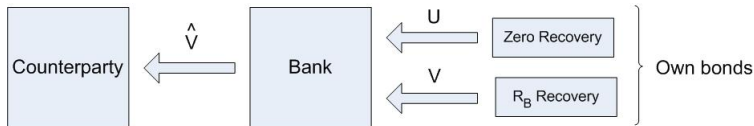
FVA in a nutshell: Derivatives liability

Funding benefit earned by bank while trade alive

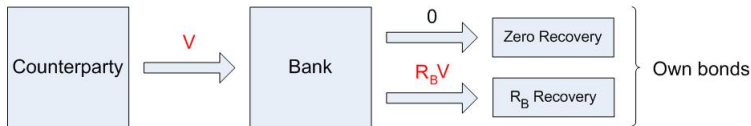
$$\begin{aligned} FBA &= - \int_t^T s_F(u) D_{r+\lambda_B+\lambda_C}(t, u) \mathbb{E}_t [V_u^-] du \\ &= -(1 - R_B) \int_t^T \lambda_B(u) D_{r+\lambda_B+\lambda_C}(t, u) \mathbb{E}_t [V_u^-] du \\ &= DVA \end{aligned}$$

- **Funding benefit is monetization of own credit risk**
- **In reality:** surplus cash will feed into bank's FTP process ...
- ... so is recycled for other funding purposes
- ... reducing overall funding needs
- ... modulo some liquidity provisions

FVA in a nutshell: Derivatives asset



Upon default:



Windfall: $(1-R_B)V^{(+)}$

FVA in a nutshell: Derivatives asset

Funding costs paid by bank while trade alive

$$\begin{aligned} FCA &= - \int_t^T s_F(u) D_{r+\lambda_B+\lambda_C}(t, u) \mathbb{E}_t [V_u^+] du \\ &= -(1 - R_B) \int_t^T \lambda_B D_{r+\lambda_B+\lambda_C}(t, u) \mathbb{E}_t [V_u^+] du \end{aligned}$$

FCA and windfall

- FCA is expectation value of windfall on negative cash account
- No monetization of windfall prior to default: include FCA in price
- Otherwise have negative drift while alive

Total adjustment

Total adjustment

$$CVA = -(1 - R_C) \int_t^T \lambda_C(u) D_{r+\lambda_B+\lambda_C}(t, u) \mathbb{E}_t [V_u^+] du$$

$$FBA = - \int_t^T s_F(u) D_{r+\lambda_B+\lambda_C}(t, u) \mathbb{E}_t [V_u^-] du$$

$$FCA = - \int_t^T s_F(u) D_{r+\lambda_B+\lambda_C}(t, u) \mathbb{E}_t [V_u^+] du$$

Can combine FCA and FBA into FVA

$$FVA = - \int_t^T s_F(u) D_{r+\lambda_B+\lambda_C}(t, u) \mathbb{E}_t [V_u] du$$

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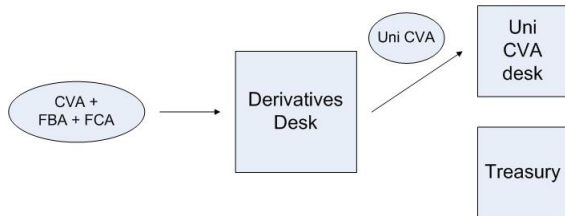
Different ways to combine

Different splits

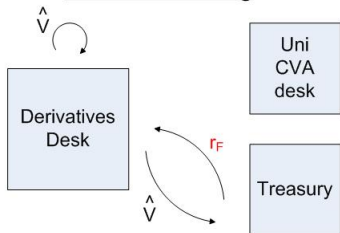
- $(CVA + DVA) + FCA = \text{bilateral CVA} + FCA$
- $CVA + (FBA + FCA) = \text{unilateral CVA} + FVA$

Setup 1: Unilateral CVA desk + Treasury

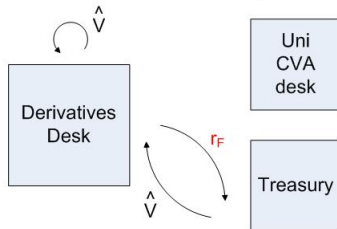
Upfront



Positive Funding

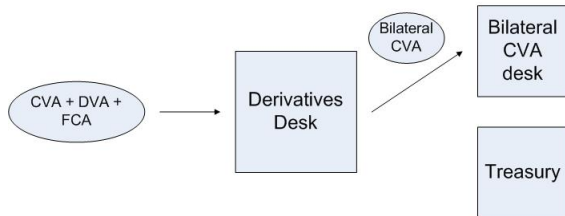


Negative Funding

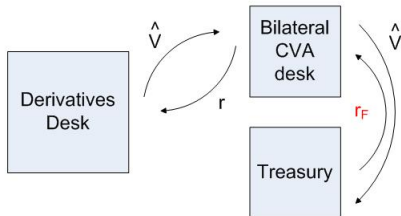


Setup 2: Bilateral CVA desk + Treasury

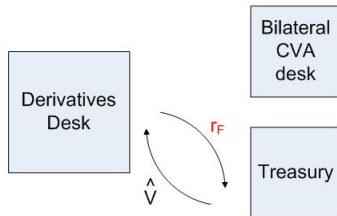
Upfront



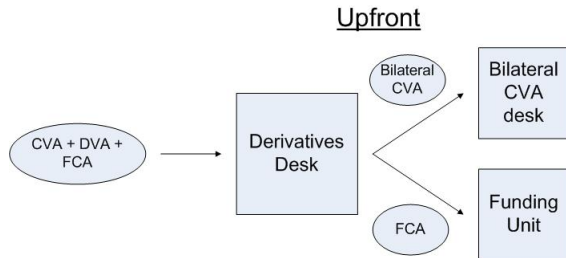
Positive Funding



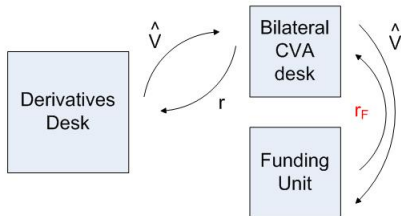
Negative Funding



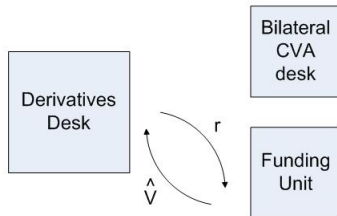
Setup 3: Bilateral CVA desk + Funding Unit



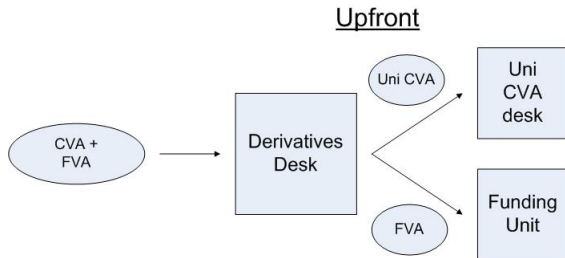
Positive Funding



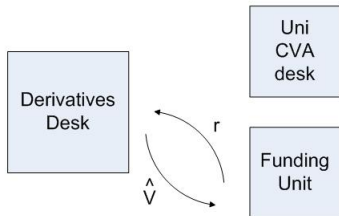
Negative Funding



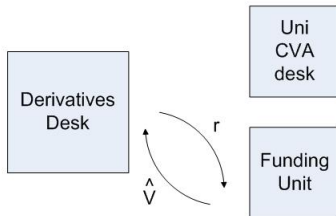
Setup 4: Unilateral CVA desk + Funding Unit



Positive Funding



Negative Funding



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General formulation - replication arguments

General close-outs

$$\begin{aligned}\hat{V}(t, S, 1, 0) &= g_B(M_B, X) && \text{Bank B defaults first,} \\ \hat{V}(t, S, 0, 1) &= g_C(M_C, X) && \text{Counterparty C defaults first,}\end{aligned}$$

- The collateral X can be a complicated process, e.g. depends on V
- $M_{B/C}$ are the close-out amounts as per CSA, e.g. ISDA
- Regular close-outs: $M_{B/C} = V$

Example: regular bilateral close-out without collateral

$$\begin{aligned}g_B &= V^+ + R_B V^- \\ g_C &= R_C V^+ + V^-\end{aligned}$$

Boundary conditions (II): Examples

Example of different close-outs

Type	$g_B(M_B, X)$	$g_C(M_C, X)$
Regular bilateral close-outs	$M_B^+ + R_B M_B^-$	$R_C M_C^+ + M_C^-$
Bilateral close-outs with collateral	$X + (M_B - X)^+ + R_B (M_B - X)^-$	$X + R_C (M_C - X)^+ + (M_C - X)^-$
Bilateral extinguisher	0	0
Bilateral setoff, without collateral	$R_B M_B$	$R_C M_C$

Semi-replication

Derivative and hedge portfolio

- The derivative $\hat{V}(t)$
- Hedge Portfolio $\Pi(t)$:
 - Market factor and counterparty risk hedges $P_C(t)$ and $S(t)$
 - $P_1(t)$ and $P_2(t)$ - two own bonds of different recoveries
 - β_S and β_C - secured funding/repo for S and P_C
 - Collateral $X(t)$

$$\Pi(t) = \delta(t)S + \alpha_1(t)P_1 + \alpha_2(t)P_2 + \alpha_C(t)P_C + \beta_S(t) + \beta_C(t) - X$$

Aim

$$\hat{V}(t) + \Pi(t) = 0 \quad \forall t < \tau_B$$

- replicate $\hat{V}$ in all scenarios but possibly issuer default

Semi-replication (ctd.)

Funding constraint

$$\hat{V} - X + \alpha_1 P_1 + \alpha_2 P_2 = 0$$

- while B is alive
- Positive cash net of collateral is invested in own bond portfolio
- Negative cash net of collateral is raised by issuing bond portfolio
- Assumes collateral re-hypothecation

Semi-replication (ctd.)

Apply usual machinery

- Ito's lemma for $d\hat{V}$
- Eliminate the stock price and counterparty risks with

$$\begin{aligned}\alpha_C P_C &= \Delta \hat{V}_C = g_C - \hat{V} \\ \delta &= -\partial_S \hat{V}\end{aligned}$$

Semi-replication (ctd.)

Portfolio evolution

$$d\hat{V} + d\tilde{\Pi} = \left(\partial_t \hat{V} + \mathcal{A}_t \hat{V} - r_X X + r_1 \alpha_1 P_1 + r_2 \alpha_2 P_2 + \lambda_C \Delta \hat{V}_C \right) dt + (g_B + P_D - X) dJ_B.$$

- $\mathcal{A}_t \equiv \frac{1}{2} \sigma^2 S^2 \partial_S^2 V + (q_S - \gamma_S) S \partial_S V$
- r_X is rate received on collateral, r_1 and r_2 are yields on P_1 and P_2
- $P \equiv \alpha_1 P_1 + \alpha_2 P_2$: pre-default value of issuer bond portfolio
- $P_D \equiv \alpha_1 R_1 P_1 + \alpha_2 R_2 P_2$: post-default value of issuer bond portfolio
- $r_i = r + (1 - R_i) \lambda_B$ - zero basis between bonds

Semi-replication (ctd.)

Simplify

Use funding constraint

$$\begin{aligned} d\hat{V} + d\tilde{\Pi} = & \left(\partial_t \hat{V} + \mathcal{A}_t \hat{V} - (r + \lambda_B + \lambda_C) \hat{V} - s_X X + \lambda_C g_C + \lambda_B g_B \right) dt \\ & - \epsilon_h \lambda_B dt + \epsilon_h dJ_B. \end{aligned}$$

- $\epsilon_h \equiv g_B + P_D - X$
- last line is a martingale
- **shareholders:** $-\epsilon_h dt$ - compensating drift term while alive
- **bondholders:** $\epsilon_h dJ_B$ - post default windfall/shortfall

Fair value and economic value

Fair value

- **Fair value** $\hat{V}_{FV}$: total value to shareholders + bondholders
- $-\epsilon_h \lambda_B dt + \epsilon_h dJ_B$ is a martingale

$$\partial_t \hat{V}_{FV} + \mathcal{A}_t \hat{V}_{FV} - (r + \lambda_B + \lambda_C) \hat{V}_{FV} - s_X X + \lambda_C g_C + \lambda_B g_B = 0$$

Economic value

- **Economic value** $\hat{V}$ to shareholders
- include $-\epsilon_h \lambda_B dt$ drift term that accrues while B is alive

$$\partial_t \hat{V} + \mathcal{A}_t \hat{V} - (r + \lambda_B + \lambda_C) \hat{V} - s_X X + \lambda_C g_C + \lambda_B g_B - \epsilon_h \lambda_B = 0$$

The general valuation adjustment PDE

Split $\hat{V}$

- $\hat{V} = V + U$
- V satisfies the regular B-S PDE
- U is total adjustment to obtain economic value

This gives PDE for U

$$\begin{aligned}\partial_t U + \mathcal{A}_t U - (r + \lambda_B + \lambda_C)U &= s_X X - \lambda_C(g_C - V) - \lambda_B(g_B - V) \\ &\quad + \lambda_B \epsilon_h \\ U(T, S) &= 0\end{aligned}$$

In this section from here: assume regular close-outs, i.e. $M = V$

- The RHS is a source term since V can be pre-computed.

Feynman-Kac solution to the valuation adjustment PDE

Integral representation for U (using Feynman-Kac)

- **Economic value adjustment** $U \equiv CVA + DVA + FCA + COLVA$ with

$$CVA = - \int_t^T \lambda_C(u) D_{r+\lambda_B+\lambda_C}(t, u) \mathbb{E}_t [V_u - g_C(V_u, X_u)] du$$

$$DVA = - \int_t^T \lambda_B(u) D_{r+\lambda_B+\lambda_C}(t, u) \mathbb{E}_t [V_u - g_B(V_u, X_u)] du$$

$$FCA = - \int_t^T \lambda_B(u) D_{r+\lambda_B+\lambda_C}(t, u) \mathbb{E}_t [\epsilon_h(u)] du$$

$$COLVA = - \int_t^T s_X(u) D_{r+\lambda_B+\lambda_C}(t, u) \mathbb{E}_t [X_u] du.$$

- **Fair value adjustment** (to bondholders and shareholders) does not include FCA.

Section summary

Take away from this section

- **Funding strategy:** $\alpha_1 P_1 + \alpha_2 P_2$
- **Own default:** windfall ϵ_h
- **Windfall due to mismatch** between
 - the uncollateralised part of the derivative close out $g_B - X$
 - and the post default own bond portfolio $\alpha_1 R_1 P_1 + \alpha_2 R_2 P_2$
- **General theorem:** FCA is discounted expectation of this hedge error
- **FCA** ensures issuer is not bleeding funding costs while alive
- For details see Burgard and Kjaer [1], [2], [3]

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Set-off close-outs

Example: Regular bilateral close-outs

For bilateral close-outs, non-defaulting party pays full amount (if it owes)

$$g_B = V^+ + R_B V^-$$

$$g_C = R_C V^+ + V^-$$

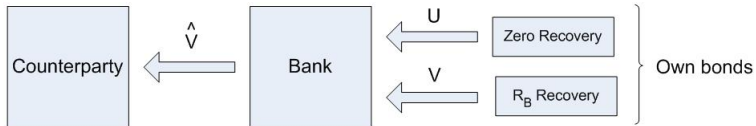
Example: Regular set-offs

For set-offs, can pay in bonds of the defaulting party, so

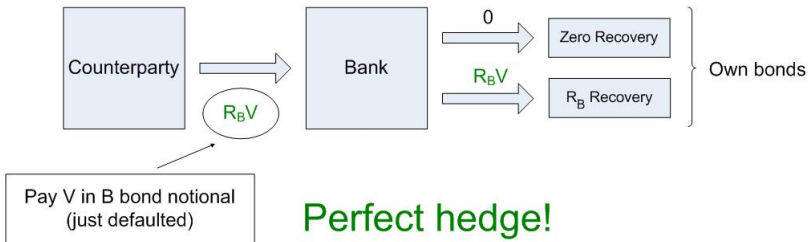
$$g_B = R_B V$$

$$g_C = R_C V$$

Derivatives asset for set-off close-outs



Upon default:



Set-off close-outs (ctd.)

So for set-offs

- No windfall
- Adjustments:

$$CVA = -(1 - R_C) \int_t^T \lambda_C(u) D_{r+\lambda_B+\lambda_C}(t, u) \mathbb{E}_t[V(u)] du$$

$$DVA = -(1 - R_B) \int_t^T \lambda_B(u) D_{r+\lambda_B+\lambda_C}(t, u) \mathbb{E}_t[V(u)] du$$

- FCA vanishes - symmetric prices

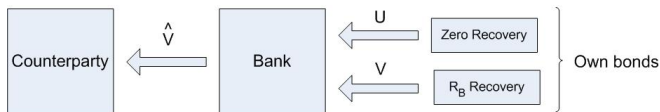
In practise:

- Hard to put set-offs into practise
- Depends on bankruptcy laws
- ... which typically don't allow this

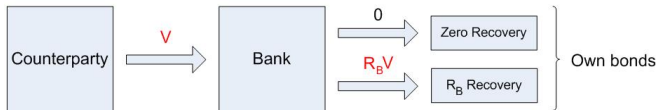
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Different funding strategies

Previous examples used a specific funding strategy (for deriv. assets)



Upon default:



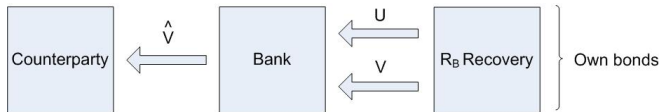
Windfall: $(1-R_B)V^{(+)}$

This gives rise to classical bilateral CVA + FCA

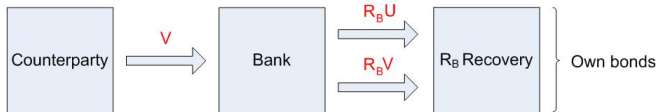
Different funding strategies (ctd.)

Can use different funding strategies

- e.g. issue single R_B bond



Upon default:



Windfall or Shortfall: $(1-R_B)V^{(+)} - R_B U$

Different funding strategies (ctd.)

For this strategy

$$\begin{aligned} FCA &= - \int_t^T \lambda_B(u) D_{r+\lambda_B+\lambda_C}(t, u) \mathbb{E}_t [\epsilon_h(u)] du \\ &= - \int_t^T \lambda_B(u) D_{r+\lambda_B+\lambda_C}(t, u) \mathbb{E}_t \left[(1 - R_B) V^+ - R_B (\hat{V} - V) \right] du \end{aligned}$$

- This is recursive ($\hat{V}$ on rhs)

Reshuffle PDE

- Can reshuffle PDE (bring $\hat{V}$ to lhs)

$$\partial_t \hat{V} + \mathcal{A}_t \hat{V} - (r_F + \lambda_C) \hat{V} = -\lambda_C g_C(V, X) - (r_F - r_X) X$$

Different funding strategies (ctd.)

Can re-shuffle PDE - get different integral representation

$$CVA_F = -(1 - R_C) \int_t^T \lambda_C(u) D_{r_F + \lambda_C}(t, u) \mathbb{E}_t [V_u^+] du$$

$$DVA_F = - \int_t^T s_F(u) D_{r_F + \lambda_C}(t, u) \mathbb{E}_t [V_u^-] du$$

$$FCA_F = - \int_t^T s_F(u) D_{r_F + \lambda_C}(t, u) \mathbb{E}_t [V_u^+] du$$

■ **Note:** FCA_F is not FCA

■ it is not the discounted expected value of the hedge error anymore

Different funding strategies (ctd.)

Conclusions funding strategies

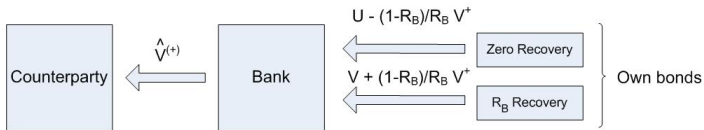
- **Different funding strategies** imply
 - **different hedge errors**
 - **different economic costs** while alive
 - **different adjustments**
- Value of the derivative including **FCA is an economic value** that reflects production costs

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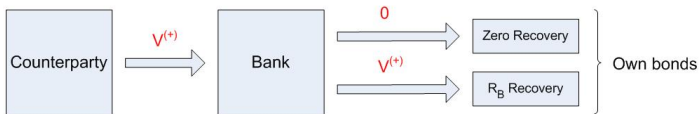
Risk neutral pricing

Theoretically, there is a strategy that hedges own default perfectly.

For derivative assets:



Upon default:



No hedge error

Risk neutral pricing (ctd.)

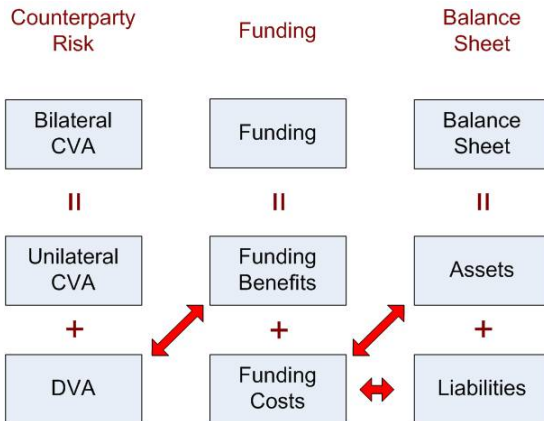
In this case

- No hedge error $\rightarrow$ no FCA
- All risks are hedged $\rightarrow$ risk neutral price
 - i.e. equivalent to discounting all derivatives cashflows at risk-free rate
- Back to classical bilateral CVA
- Effectively, fine tuning of junior vs senior bonds to match own-default profile of derivatives portfolio

In practice

- **Other constraints** on balance sheet to prevent this
- Would require **significant repurchase of own junior bonds**
- Would likely **break bond covenants**
- In general **not outstanding in that volume** to start with

Funding and balance sheet



Funding and balance sheet (CTD.)

Derivatives, balance sheet and funding spread

- **Derivatives and funding:** impact balance sheet
- **Balance sheet:** impacts funding spreads
- **Funding spreads:** impact derivative funding

Funding and balance sheet (CTD.)

Simple balance sheet model (see Burgard and Kjaer [2])

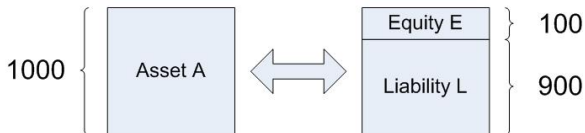
- Simple Merton type model for default
- Add derivative asset and funding liability
- Recovery rate changes
- Feedback into funding spread
- If old debt is floating credit, then marginal funding cost is 0

Does this argument work in reality?

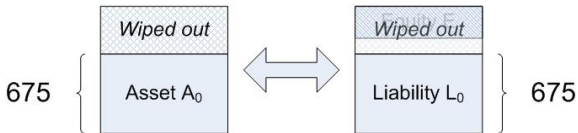
- This balance sheet model is simplistic
- In reality: funding is driven by many other factors
- ... and a significant portion of existing funding is at fixed rate

Case Study Balance Sheet: asset and funding impact

Balance Sheet:



Asset induced default – with 10% probability



Liability Recovery :

$$R = 675/900 = 75\%$$

Case Study Balance Sheet: asset and funding impact

Context

- Assume $r=0\%$
- Asset drop: 10% probability for a 32.5% drop
- Asset earns fair spread of 3.25 %
 - no risk premium
- So induced bank default:
 - Induced default probability: 10%
 - Induced bank recovery rate: $R=75\%$
- Implied bank funding spread: $s_F = 2.5\%$

Case Study Balance Sheet (ctd.)

Case study

Compare base case with

- Add derivative asset (100)
- and corresponding debt
- Compare: if we do not add FCA
 - for floating debt
 - for fixed debt
- To: if we do add FCA
 - for fixed debt

Case Study Balance Sheet (ctd.)

	Base case	Floating credit no FCA	Fixed credit no FCA	Fixed credit with FCA
Asset	1000	1100	1100	1097.53
Debt	-900	-1000	-1000	-997.53
Equity	-100	-100	-100	-100
Equity Ratio	10%	9.09%	9.09%	9.11%
R_B	75.00%	77.50%	77.50%	77.69%
$r_F^{old_debt}$	2.50%	2.25%	2.50%	2.50%
$r_F^{new_debt}$	-	2.25%	2.50%	2.50%
Asset income	33.03	33.03	33.03	33.03
Interest costs	-22.78	-22.78	-25.32	-25.25
Net income	10.25	10.25	7.72	7.78
Asset	1010.25	1110.25	1107.72	1107.78
Debt	-900	-1000	-1000	-997.53
Equity	-110.25	-110.25	-107.72	-110.25
RoE	9.76%	9.76%	7.44%	9.76%

Case Study Balance Sheet (ctd.)

	Base case	Floating credit no FCA	Fixed credit no FCA	Fixed credit with FCA
Dividend	-10.25	-10.25	-7.72	-10.25
Asset	1000	1100	1100	1100
Debt	-900	-1000	-1000	-1000
Equity	-100	-100	-100	-100
Equity Ratio	10%	9.09%	9.09%	9.09%
<i>sell derivative, pay back debt</i>				
Asset	1000	1000	1000	1000
Debt	-900	-900	-900	-900
Equity	-100	-100	-100	-100
Equity Ratio	10%	10%	10%	10%

Case Study Balance Sheet (ctd.)

If bank were not to charge FCA

- **If existing debt is at fixed rate**

- balance sheet feedback effect doesn't help
- Even if *new* funding for derivatives position is at new rate, this has only marginal impact
- Own credit adjustment on existing debt does not help either
- In fact, it would just bring the loss forward

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Accounting for FVA

Are you an accountant?

- No

Has any bank started accounting for FVA?

- In 2012 results, four banks have started to account for FVA
 - RBS, Barclays, GS, LLoyds
- 2013 has seen some more
 - JPM, DB, Nomura

Is there a market consensus?

- Market consensus is still evolving
- But there seems to be general agreement that funding costs should be accounted for

Accounting for FVA (ctd.)

Which funding spread to use?

- For accounting purposes, assets valued at realizable prices
- So accountants like to use a "market funding spread" rather than idiosyncratic
- E.g. JPM seems to have accounted for a "market funding spread" of 50bps - with idiosyncratic one around 70bps
- How to determine a "market funding spread"?
- Still, economic cost encountered will be driven by the idiosyncratic spread
- So realised economic value of a trade may not accrue to the accounted value

Accounting for FVA (ctd.)

Is the economic value or the fair value accounted for?

- Can be both (see Albanese and Andersen [4])

Are FCA, FBA, FVA on netting sets, funding sets or bank-wide?

- Realised economic value depends on the funding strategy
- For the strategies presented above, FCA and FBA are calculated on netting sets ...
- ... and they are additive across netting sets
- $FVA = FCA + FBA$ is linear, so does not depend on how it's added
- There are other strategies, where FCA and FBA are non-additive
- e.g. investing surplus cash risk-free, see Albanese and Andersen [4]

How to move from bilateral CVA to FVA?

- Either
 - Take away DVA
 - Add FVA
- Or (equivalently)
 - Add difference $FBA(\text{market funding spread}) - DVA(\text{CDS spread})$
 - Add FCA (market funding spread)

Case study

Case study: bank with

- DVA = USD 2.5 bln
- cds spread 65 bps
- market funding spread 50 bps
- DVA (structured notes and derivatives) of USD 2.5bln
- uncollateralised derivatives payables USD 50 bln with 5 years average lifetime
 - added over netting units

Case study (ctd.)

Case study: back of the envelope accounting

- DVA to FBA:
 - move from CDS to market funding spread
 - so scale down by 50bps/65bps
 - loss of USD 577 mio
- FCA:
 - cost of roughly $\text{USD } 50 \text{ bln} * 50 \text{ bps/yr} * 5 \text{ yr} = \text{USD } 1250 \text{ mio}$
- Overall loss of about USD 1.8 bln

Accounting for FVA (ctd.)

Will results fluctuate wildly with funding spread once on FVA?

- No, not really
- In fact, funding spread sensitivity smaller than for bilateral CVA
- FCA has opposite sign to FBA/DVA, so expect some cancellation
- But what about CVA then?
 - systemic risk and some idiosyncratic can be hedged out

Accounting for FVA (ctd.)

Still sensible to hedge DVA/FBA by selling protection on basket of financials?

- Depends
- It's not really a hedge against idiosyncratic risk of own default
- Could be used to hedge remaining FVA risk to market funding spread
- But induces noise from defaults in the basket
- In the end it's a trading strategy - earn premium and pay when there is a default in the basket

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Conclusions

- **Funding and DVA** are intrinsically linked
- **Different funding strategies**
 - ... imply different funding costs
 - ... and economic values for derivatives
- **FCA**
 - ... depends on funding strategy employed
 - ... corresponds to expected hedge error upon own default
 - ... ensures bank shareholders don't lose money while bank alive
- **Balance sheet and funding**
 - ... are closely linked
 - ... relationship and feedback effects not straight forward in practice
 - ... strong feedback effects only if credit were floating

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